Control work 8 · Annual review

The whole of Grade 8 algebra in one paper — and the four habits that decide the grade.

LESSON 101–102 | 2 × 40 MIN | ALGEBRA 8 — ANNUAL REVIEW | STAGE 9 · UNITS 1–15

LESSON CLOCK

2' | 36' | 2'

Setting up	2 min
The paper	36 min
Collect in	2 min

LEARNING OBJECTIVES

- Assess the year: fractions, roots, quadratics, inequalities, data and counting.
- Diagnose each lost mark by error type.
- Set out the summer work that follows from the result.

TEXTBOOK

National	Chapters I–IV
Cambridge	Learner's Book, whole course
Workbook	Workbook, whole course

01 Explanation

Lesson 101 — the paper (40 minutes)

Two variants of ten tasks. Tasks 1–6 are worth 1 mark, tasks 7–10 are worth 2:

- task 1 · simplify an algebraic fraction
- task 2 · add two fractions with different denominators
- task 3 · simplify a square root
- task 4 · solve a quadratic equation
- task 5 · solve a linear inequality and show it on a number line
- task 6 · find the mean, median and mode of a small list
- task 7 · a word problem leading to a quadratic
- task 8 · a system of two inequalities
- task 9 · the nth term or the gradient of a line
- task 10 · a counting or probability question

Marking note: in task 1 the excluded values must be stated; in task 7 the impossible root must be rejected in writing.

Lesson 102 — work on mistakes and the summer plan (40 minutes)

THE FOUR HABITS THAT DECIDE THIS PAPER

1. **Write the domain** before touching an algebraic fraction.
2. **Never divide by an unknown** — factorise, so that $x = 0$ is not lost.
3. **Reverse the inequality sign** whenever you multiply or divide by a negative number.
4. **Check the answer against the question**, not just against your own working — a length cannot be negative, a probability cannot exceed 1.

Each pupil writes the habit number beside every lost mark. The topics that appear twice in that list become the summer plan: five problems a week from that section, no more.

LOOKING AHEAD TO GRADE 9

Everything next year rests on three things from this one: factorising confidently, solving a quadratic without hesitation, and reading a graph. If any of those is shaky, that is where the summer goes.

02 Worked examples

EXAMPLE 1 **Find the error: $\frac{x^2-9}{x-3} = x + 3$ for every x**

1. The cancelling is right, but the domain is not.
   $x - 3$ is in the denominator.

2. $x \neq 3$
   At $x = 3$ the original fraction has no value at all.

3. $\frac{x^2-9}{x-3} = x + 3, x \neq 3$
   The restriction is part of the answer.

ANSWER

$x + 3, x \neq 3$

EXAMPLE 2 **Find the error: $-2x > 6$ so $x > -3$**

1. Both sides were divided by -2.
   That reverses the order.

2. $x < -3$
   The sign turns round.

3. $x = -5$: $-2 \cdot (-5) = 10 > 6$ ✓
   Test a value to be sure.

ANSWER

$x < -3$

03 Interactive model

Show each piece of wrong working, take a vote on the habit that was missing, then reveal.

Which habit was missing?

Claimed: $\dfrac{x^2 - 4}{x - 2} = x + 2$ for every x

1. $x^2 - 4 = (x - 2)(x + 2)$

   The factorising is correct.

2. The denominator vanishes at $x = 2$.

   There the original expression is undefined.

3. Write $x \neq 2$ beside the answer.

   A mark hangs on that line.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Annual review	Yillik takrorlash	Годовое повторение
Control work	Nazorat ishi	Контрольная работа
Domain (allowed values)	Aniqlanish sohasi	Область допустимых значений
Discriminant	Diskriminant	Дискриминант
Inequality	Tengsizlik	Неравенство
Modulus	Modul	Модуль
Estimated mean	Taxminiy oʻrta	Оценка среднего
Counting principle	Sanashning asosiy qoidasi	Основной принцип счёта
Check by substitution	Oʻrniga qoʻyib tekshirish	Проверка подстановкой

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$\frac{x^2 - 25}{x - 5}$ simplifies to:

- $x + 5$
- $x + 5, x \neq 5$
- $x - 5$
- $x^2 - 5$

$-4x < 20$ gives:

- $x < -5$
- $x > -5$
- $x < 5$
- $x > 5$

$2x^2 = 10x$ has roots:

- $x = 5$
- $x = 0, 5$
- $x = \pm 5$
- $x = 0$

A probability comes out as 1.2. This means:

- the event is certain
- there is an error
- the event is very likely
- multiply by 100

$\sqrt{50}$ in simplest form is:

- $5\sqrt{2}$
- $2\sqrt{5}$
- $25\sqrt{2}$
- $10\sqrt{5}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. ***Variant 1, task 1.*** *Simplify* $\frac{x^2 - 16}{x + 4}$ *and state the excluded value.*

   ANSWER $x - 4, x \neq -4$

2. ***Task 2.*** *Simplify* $\frac{1}{x} + \frac{1}{2x}$

   ANSWER $\frac{3}{2x}, x \neq 0$

3. ***Task 3.*** *Simplify* $\sqrt{72}$

   ANSWER $6\sqrt{2}$

4. ***Task 4.*** *Solve* $x^2 - 7x + 10 = 0$

   ANSWER $x = 2, x = 5$

5. ***Task 5.*** *Solve* $3x - 4 < 11$

   ANSWER $x < 5$

6. ***Task 6.*** *Find the mean, median and mode of 4, 8, 4, 6, 8, 4*

   ANSWER *mean 5.67, median 5, mode 4*

7. ***Task 9.*** *Find the nth term of 5, 9, 13, 17, …*

   ANSWER $4n + 1$

Medium · the standard question

7 PROBLEMS

1. ***Variant 2, task 1.*** *Simplify* $\frac{3x + 12}{x^2 - 16}$

   ANSWER $\frac{3}{x - 4}$, $x \neq \pm 4$

2. ***Task 2.*** *Simplify* $\frac{2}{x - 1} - \frac{1}{x}$

   ANSWER $\frac{x + 1}{x(x - 1)}$, $x \neq 0, 1$

3. ***Task 4.*** *Solve* $2x^2 - 7x + 3 = 0$

   ANSWER $x = 3, x = 0.5$

4. ***Task 5.*** *Solve* $5 - 2x \geq 1$ *and show it on a number line.*

   ANSWER $x \leq 2$

5. ***Task 8.*** *Solve the system* $x + 1 > 0$ *and* $2x - 6 < 0$.

   ANSWER $-1 < x < 3$

6. ***Task 9.*** *Find the gradient of the line through (1, 4) and (5, 16).*

   ANSWER 3

7. ***Task 10.*** *Two dice are thrown. Find P(total 6).*

   ANSWER $\frac{5}{36}$

Hard · stretch and reason

7 PROBLEMS

1. ***Task 7.*** *A rectangle is 4 cm longer than wide and has area 96 cm².* *Find its sides.*

   ANSWER 8 cm × 12 cm (reject -12)

2. ***Task 7, variant 2.*** *A number and its square add to 56. Find the positive value.*

   ANSWER 7

3. *Solve* $|x - 3| < 5$

   ANSWER $-2 < x < 8$

4. *Solve* $x^2 - 5x + 6 > 0$

   ANSWER $x < 2$ or $x > 3$

5. *A code of 3 digits from 1–5 must contain at least one 5. How many codes?*

   ANSWER $125 - 64 = 61$

6. *Simplify* $\dfrac{x^2 - 5x + 6}{x^2 - 9}$ *and state the excluded values.*

   ANSWER $\dfrac{x - 2}{x + 3}$, $x \neq \pm 3$

7. *One root of* $x^2 + bx + 21 = 0$ *is 3. Find b and the other root.*

   ANSWER $b = -10$, other root 7

07 Homework

Summer work — 5 tasks

Not a punishment: twenty minutes a week keeps Grade 9 easy.

1. Re-solve, in full, each task you lost a mark on, with the habit number (1–4) beside it.
2. List the two topics that appear most often in that list — those are your summer topics.
3. Five problems a week from each of those two topics. No more.
4. *Practise until you can factorise* $x^2 + bx + c$ *and solve any quadratic without hesitating.*

5. *Practise reading the gradient and intercept off any straight-line graph.*